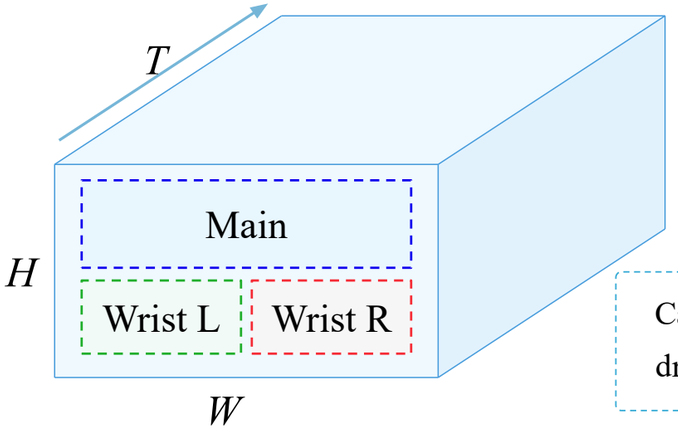
INPUT STREAMS

Main RGB | Wrist L RGB | Wrist R RGB | Depth | Segmentation | Edge

$H_i \times W_i \times T$

VAE → PatchEmbed → ⊕ ← View + modality → …

VIRTUAL CANVAS



Canvas coords drive 3D RoPE

Views: separate spatial slots

PACKED SEQUENCE

	Main								Wrist L				Wrist R				Depth			Seg.			Edge		
τ Noise	0	σ	σ	σ	σ	σ	σ	σ	0	σ	σ	σ	σ	σ	σ	σ	0	0	0	σ	σ	σ	σ	σ	σ
View ID	0	0	0	0	0	0	0	0	1	1	1	1	2	2	2	2	0	0	0	0	0	0	0	0	0
Modality ID	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	2	2	2	3	3	3
RoPE coords	π_i	π_i	…					π_i	π_j	…			π_k	…			π_i	…	π_i	π_i	…	π_i	π_i	…	π_i
Patch tokens																									

VLM

DiT

Per-token modulation (τ)

AdaLN

Self-attn + 3D RoPE

Cross-attention

AdaLN

Feed-forward

× N blocks

decode

OUTPUT STREAMS

Main RGB | Wrist L RGB | Wrist R RGB

Depth | Seg. | Edge

$\tau = 0$ (clean) $\tau = \sigma$ (noisy) ⊕ ADD